

Transpupillary Thermotherapy versus Plaque Radiotherapy for Suspected Choroidal Melanomas

J. William Harbour, MD,¹ Travis A. Meredith, MD,² Paul A. Thompson, PhD,³ Mae E. Gordon, PhD³

Purpose: To examine the outcomes and complications of transpupillary thermotherapy for treatment of small choroidal melanomas.

Design: Retrospective case-matched comparative study and retrospective observational study.

Participants: The case-matched study consisted of 36 patients treated with either transpupillary thermotherapy or plaque radiotherapy (18 patients per group). The observational study consisted of 21 patients treated with transpupillary thermotherapy alone (primary transpupillary thermotherapy) and 9 patients treated with transpupillary thermotherapy plus plaque radiotherapy (adjuvant transpupillary thermotherapy).

Main Outcome Measures: Visual acuity, local tumor control, and metastasis.

Results: The case-matched groups did not differ significantly with respect to age, gender, initial visual acuity, tumor location, or length of follow-up (mean, 33 months for transpupillary thermotherapy vs. 40 months for plaque radiotherapy). There was no significant difference in final visual acuity ($P = 0.810$) or postoperative visual acuity change ($P = 0.919$) between the 2 groups. In the observational study, the mean follow-up was 32 months (range, 4–58 months). Indications for primary transpupillary thermotherapy included documented growth (10 patients) and the presence of ≥ 3 risk factors for growth (11 patients). Retinal complications occurred in 16 patients (76%). The mean posttreatment visual acuity change was minus 2 lines (range, minus 9 to plus 7 lines). Local failure occurred in 6 patients (29%). Local failure was associated with an increased number of transpupillary thermotherapy spots per session ($P = 0.023$) and decreased tumor pigmentation ($P = 0.001$). Indications for adjuvant transpupillary thermotherapy included tumor radioresistance (6 patients) and the presence of risk factors for local failure (3 patients). All 9 tumors that received adjuvant transpupillary thermotherapy regressed rapidly, with no local failures. The mean postoperative visual acuity change was -1.9 lines (range, -9 to $+5$ lines). No patient in either study developed metastasis.

Conclusions: The recent interest in transpupillary thermotherapy as primary therapy for choroidal melanoma is based largely on the assumption that transpupillary thermotherapy may provide better visual outcomes than plaque radiotherapy. However, this study found no significant difference in visual outcomes between transpupillary thermotherapy and plaque radiotherapy. Further, the local failure rate with transpupillary thermotherapy was substantially higher than with plaque radiotherapy. The most promising role for transpupillary thermotherapy seems to be as an adjunct to plaque radiotherapy. The appropriate indications for transpupillary thermotherapy in the management of choroidal melanoma need to be re-evaluated. *Ophthalmology* 2003;110:2207–2215 © 2003 by the American Academy of Ophthalmology.

Choroidal melanoma is the most common primary malignancy of the eye and is often treated by plaque radiotherapy.¹ Although plaque radiotherapy has an excellent local

control rate, radiation complications can significantly compromise visual acuity, especially for tumors near the macula or optic disc. In an attempt to reduce treatment-related vision loss in small posterior melanomas, a number of centers have advocated the use of transpupillary thermotherapy (TTT), which consists of the delivery of a long-duration, long-wavelength (810 nm) diode laser through a large spot size.²

Although TTT is now widely used for treating choroidal melanomas, the appropriate indications for this modality remain unclear. Transpupillary thermotherapy was originally developed by Oosterhuis et al^{2,3} as an adjunct to plaque radiotherapy. These workers expressed reservations about the use of TTT alone without plaque radiotherapy because TTT did not eradicate intrascleral tumor cells,⁴ which may be present in up to 55.7% of uveal melanomas.⁵ Most other European centers have also emphasized the use

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¹ Ocular Oncology Service, Department of Ophthalmology and Visual Sciences, Washington University School of Medicine, St. Louis, Missouri.

² Department of Ophthalmology, University of North Carolina, Chapel Hill, North Carolina.

³ Division of Biostatistics, Washington University School of Medicine, St. Louis, Missouri.

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Correspondence to J. William Harbour, MD, Campus Box 8069, Washington University School of Medicine, 660 South Euclid Avenue, St. Louis, MO 63110. E-mail: harbour@vision.wustl.edu.

of TTT as an adjunct to plaque radiotherapy.^{6,7} Nevertheless, a number of centers in the United States have used TTT alone as primary treatment for small posterior melanomas.^{8–10} The initial reports from these centers were promising, with low failure rates (0%–7%) and favorable visual outcomes. However, the follow-up for these studies was very short (mean, 7–18 months). With longer follow-up, most centers have observed higher local failure rates, as demonstrated by 3 consecutive reports from one group.^{10–12}

In addition to increasing local failure rates, most centers have encountered a variety of vision-threatening complications with TTT, including macular pucker, cystoid macular edema, retinal vascular occlusion, macular traction and distortion, vitreous and subretinal hemorrhage, exudative and rhegmatogenous retinal detachment, vitreitis, and intravitreal pigment dispersion.^{8,11,13,14} Because a major goal of primary TTT is to reduce treatment-related vision loss, the high rate of complications raises the question of whether visual outcomes with TTT are truly superior to those with plaque radiotherapy. Unfortunately, no comparative studies exist to address this question. Therefore, we have performed a case-matched comparison of patients treated with TTT vs. plaque radiotherapy and have analyzed the visual outcomes and tumor control in our series of choroidal melanoma patients treated with TTT. This study represents the longest follow-up reported to date.

Materials and Methods

Patient Selection and Treatment

The study was performed in accordance with the institutional review board of Washington University. A computer search of the Barnes Retina Institute records identified 30 patients with choroidal melanomas or choroidal melanocytic tumors suspected of being small melanomas that were treated with TTT as primary therapy or as an adjuvant to iodine 125 plaque radiotherapy between March 1997 and November 2001 by one of the authors (JWH or TAM). Patients were excluded if TTT was performed at another institution (2 patients), if TTT was performed as a nonablative procedure to reduce subretinal fluid in a premalignant lesion (one patient), or if the treatment technique consisted of standard laser photocoagulation rather than TTT (one patient). The following clinical data were recorded: patient age, gender, race, involved eye, degree of tumor pigmentation (1, minimal; 2, moderate to heavy), and location of the tumor epicenter with respect to the fovea (temporal, nasal, inferior, or superior). Proximity to the optic nerve and fovea was also noted: juxtapapillary ($\leq 200 \mu\text{m}$ from the optic disc edge), juxtafoveal ($\leq 200 \mu\text{m}$ from the foveal avascular zone), subfoveal, or juxtamacular ($\leq 200 \mu\text{m}$ from the temporal vascular arcades). Transpupillary thermotherapy was used to treat posterior tumors near the macula, optic disc, or both with thickness $\leq 3.5 \text{ mm}$ (there was one exception: height, 4.2 mm).

Indications for TTT included the following: (1) documented tumor growth of $\geq 0.2 \text{ mm}$ in horizontal or vertical dimension (smaller changes were usually verified by repeat examination); (2) presence of 3 or more risk factors for growth, including subretinal fluid, orange pigmentation, thickness $> 2 \text{ mm}$, and visual symptoms (most of these tumors were close to the fovea, where waiting for documented growth may have resulted in tumor extension into the fovea); (3) tumor unresponsiveness to plaque radiotherapy; or

(4) prophylactic adjuvant therapy after plaque radiotherapy in a tumor with high-risk factors for local failure (e.g., juxtapapillary tumor location or plaque tilting away from the sclera detected by intraoperative ultrasound). Snellen visual acuity and the following tumor features were recorded before surgery and at each postoperative visit: largest basal tumor diameter and tumor thickness (by standardized A-scan ultrasonography). The following postoperative events were recorded: tumor flattening (by ultrasonography), treatment complications, local treatment failure (tumor persistence or recurrence requiring further treatment besides TTT), treatment for local failure (plaque radiotherapy or enucleation), and systemic metastasis. Transpupillary thermotherapy parameters were recorded for each treatment session, including date, number of TTT spots, treatment power (milliwatts), and spot size.

Transpupillary thermotherapy was performed as previously described.² A 3-mm spot size was used for most cases, although a smaller spot size was occasionally used for smaller tumors or for treatments near the optic disc or fovea. Overlapping spots were applied until the entire tumor surface was treated, including 200 to 400 μm of normal choroid at the tumor margins. A smaller margin was treated near the fovea. Adjuvant TTT was administered 3 months after plaque radiotherapy. Patients were followed up at least every 3 months for the first year, at least every 4 months for the second year, and at least every 6 months thereafter. After the initial session, additional TTT was performed only if there was residual tumor tissue, as reflected by persistent pigmentation (by ophthalmoscopic examination), intrinsic vascularity (by fluorescein angiography), or tumor elevation (by ultrasonography).

Patient Matching

Patients treated with TTT alone were matched with plaque radiotherapy patients by using a computer search to identify all patients treated with plaque radiotherapy at the Barnes Retina Institute between September 1992 and May 2000. Patients were matched first on the basis of initial visual acuity, second on tumor location (juxtapapillary, nasal, macular, juxtamacular, or subfoveal), and third on preoperative tumor thickness. Three TTT patients with subfoveal tumors were excluded because their initial visual acuity was worse than that of any of the plaque radiotherapy patients available for analysis. The following factors were recorded for each group: age, gender, basal tumor diameter, tumor thickness, tumor location, initial and final visual acuity, follow-up in months, local treatment failure, and systemic metastasis.

Statistical Analysis

To analyze visual outcomes, Snellen visual acuities were converted to a nonlinear scale from 0 to 1 that was obtained by dividing the numerator by the denominator (e.g., 20/20 = 1.0; 20/400 = 0.05). Counting fingers, hand movements, and no light perception (including enucleated eyes) were assigned values of 0.02, 0.01, and 0, respectively. Change in lines of visual acuity was calculated from Snellen lines (i.e., one line each for 20/15, 20/20, 20/25, 20/30, 20/40, 20/50, 20/60, 20/70, 20/80, 20/100, 20/200, and 20/400). Vision of counting fingers, hand movements, and no light perception/enucleation were each assigned a value of 1 line. Visual outcomes were compared in several ways: (1) mean post-treatment final visual acuity (2-tailed Student's *t* test), (2) mean posttreatment change in visual acuity measured in Snellen lines (2-tailed Student's *t* test), (3) proportion of patients with final visual acuity of 20/400 or worse (chi-square test), and (4) proportion of patients with final visual acuity the same or better than initial visual acuity (chi-square test). The time dependency of visual changes between the 2 groups was analyzed by life-table analysis of the time to visual failure, defined as final visual acuity

Table 1. Analysis of Clinical Outcomes after Primary Transpupillary Thermotherapy (21 Patients)

Clinical Factor	Clinical Outcome Measures					
	Visual Acuity Change		Retinal Complications		Local Failure	
	r	P	r	P	r	P
Age	−0.280	0.092	0.070	0.382	0.664	0.257
Female	−1.893	0.074	1.652	0.116	0.271	0.791
Largest basal diameter	−0.234	0.253	0.082	0.362	1.183	0.252
Thickness	−0.284	0.138	0.111	0.316	1.512	0.148
Superior location	0.047	0.963	1.542	0.138	−0.399	0.694
Temporal location	0.208	0.837	−1.791	0.090	1.372	0.186
Nasal location	−0.337	0.739	0.082	0.936	−1.163	0.258
Juxtapapillary location	−0.838	0.412	−0.647	0.526	−0.399	0.694
Pigment	1.250	0.226	1.859	0.078	−3.943	0.001
Total spots	−0.526	0.007	0.219	0.170	1.494	0.152
Tumor flattening	0.307	0.762	2.435	0.024	−1.032	0.316
Spots per session	−0.506	0.002	0.027	0.444	2.253	0.036
Number of sessions	−0.206	0.019	0.250	0.137	0.662	0.516
Maximum power	−0.658	0.001	0.258	0.129	1.843	0.082

P = P value; r = Pearson correlation coefficient.

≤20/400. The statistical power of the study was determined by calculating the number of patients in each group that would be required to detect a significant difference in visual outcomes, assuming a 2-sided P value of 0.05, a statistical power of 80%, and constancy of other variables.

The following treatment parameters and outcomes were analyzed for association among patients treated with primary TTT: visual acuity change (Snellen lines), retinal complications (number per patient), local treatment failure (present or absent), total TTT spots, mean number of spots per session, number of sessions, and maximum power (milliwatts). Pearson correlation coefficients were used to analyze continuous, quantitative variables. Two-tailed Student's t tests were used to analyze distributions of continuous variables. For all statistical calculations, probability values were obtained from the corresponding 2-tailed Student's t value. The threshold of significance was set at $P < 0.05$.

Results

Thirty patients met inclusion criteria for the observational study, including 21 males and 9 females. The mean age was 63.7 years (range, 36–82 years). Initial treatment was primary TTT in 21 patients and adjuvant TTT in 9 patients.

Primary TTT

For the 21 patients initially treated with TTT alone, the mean pretreatment visual acuity was 20/167 (range, 20/15 to 20/1000), and the mean final visual acuity was 20/700 (range, 20/20 to hand movements). The mean change in lines of visual acuity was minus 2 lines (range, minus 9 to plus 7 lines). The tumor location was juxtapapillary (9 patients), juxtafoveal or subfoveal (5 patients), or juxtamacular (7 patients). The fundus quadrant involved was superior (12 patients), nasal (3 patients), or temporal (6 patients). The mean largest basal tumor diameter was 7.3 mm (range, 4–13 mm), and the mean tumor thickness was 2.6 mm (range, 1.8–4.2 mm). The indications for TTT were documented tumor growth (10 patients) or the presence of ≥3 risk factors for growth (11 patients). The mean number of TTT sessions was 2.4 (range, 1–4). The tumor flattened by ultrasonography in 17 patients (81%), with

a mean time to flattening of 14 months (range, 2–42 months; Fig 1). Fluorescein angiography was helpful in detecting persistent marginal tumor expansion (Fig 2) and local tumor recurrence (Fig 3). The mean number of TTT spots per session was 15 (range, 8–26), and the mean total number of TTT spots was 36.2 (range, 8–62 spots). The mean maximum treatment power was 730 mW (range, 350–1500 mW). Complications included macular pucker (9 patients), branch retinal vein occlusion (6 patients), cystoid macular edema (5 patients), vitreous and/or retinal hemorrhage (3 patients), branch retinal artery occlusion (2 patients), rhegmatogenous retinal detachment (2 patients), and retinal neovascularization (one patient). At least 1 complication occurred in 16 patients (76%). Two tumors did not respond to TTT, and 3 tumors recurred after initial flattening. The mean time to local recurrence was 32 months (range, 6–49 months). Five (24%) of the persistent or recurrent tumors were considered local failures and were managed by enucleation (3 patients) or plaque radiotherapy (2 patients). After a mean follow-up of 32 months (range, 4–58 months), ocular salvage was achieved in 18 (86%) patients. Residual pigmentation was present within the tumor scar in 25 (93%) of 27 retained eyes. No patients developed metastatic disease.

Analysis of Treatment Parameters and Outcomes for Primary TTT

Clinical variables were analyzed for association with TTT outcomes (Table 1). Patients treated with TTT alone ($n = 21$) were used for this analysis. Decreased final visual acuity was associated with increased total number of TTT spots ($P = 0.007$), increased spots per session ($P = 0.002$), increased number of TTT sessions ($P = 0.019$), and increased maximum TTT power ($P = 0.001$). The number of retinal complications was associated with flattening of the tumor ($P = 0.024$). Local treatment failure was associated with increased TTT spots per session ($P = 0.036$) and decreased tumor pigmentation ($P = 0.001$).

Clinical variables were analyzed for association with TTT treatment parameters (Table 2). Total number of TTT spots was associated with increasing larger basal tumor diameter ($P = 0.045$). The number of treatment spots per session was associated with increased tumor thickness ($P < 0.001$), increased largest basal tumor diameter ($P < 0.001$), and advanced patient age ($P = 0.044$). Maximum treatment power was associated with increased tumor

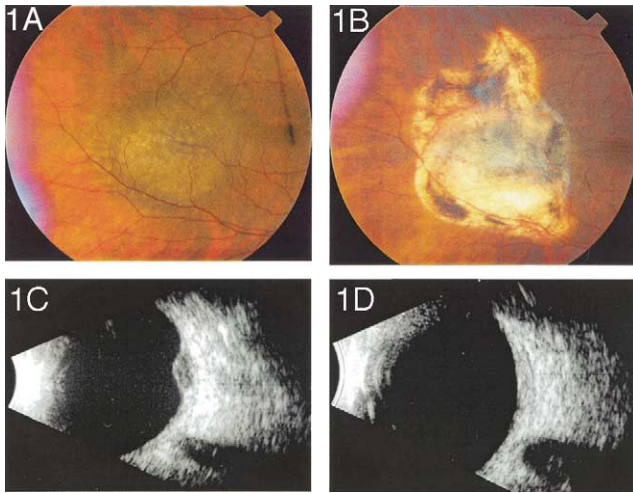


Figure 1. Example of successful primary transpupillary thermotherapy (TTT). **A**, Pretreatment tumor appearance. **B**, Sharp demarcation of treatment scar from surrounding choroid 17 months after one TTT treatment. Note the scattered residual pigmentation and overlying fibrous metaplasia. **C**, Pretreatment B-scan ultrasonography, showing an elevated choroidal tumor. **D**, B-scan ultrasonography 17 months after TTT, showing complete flattening of the tumor and no extrascleral tumor extension.

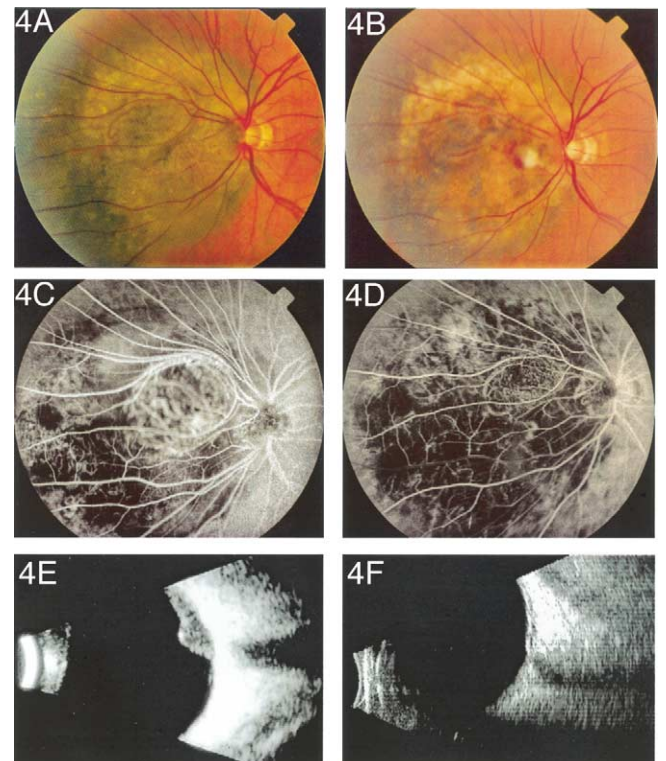
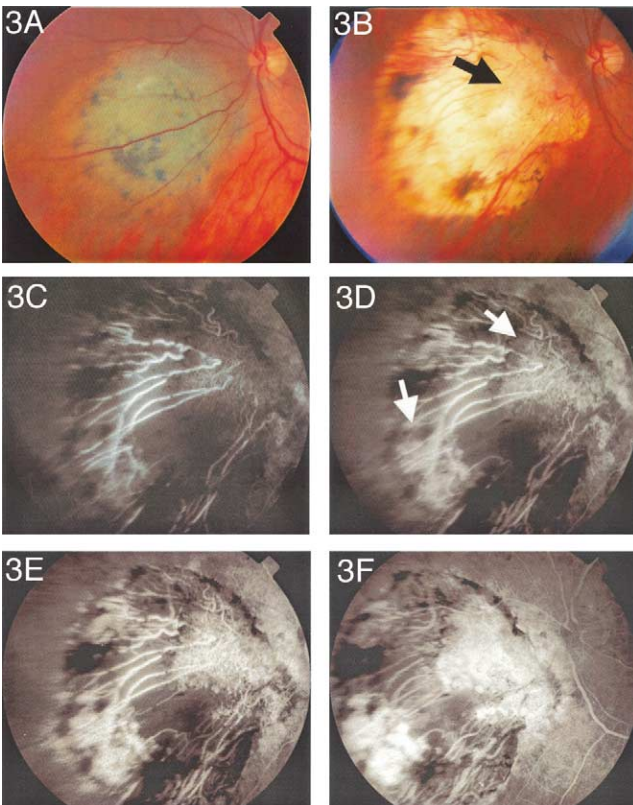
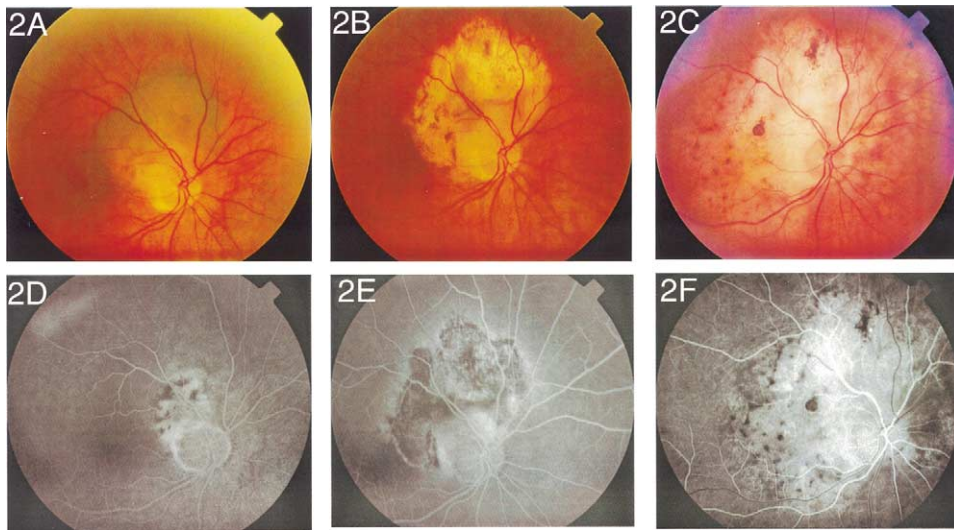


Table 2. Analysis of Treatment Parameters for Primary Transpupillary Thermotherapy (TTT; 21 Patients)

Clinical Factor	TTT Parameters							
	Total Spots		Spots per Session		No. of Sessions		Maximum Power	
	r	P	r	P	r	P	r	P
Age	-0.076	0.372	0.316	0.044	-0.313	0.084	0.375	0.047
Female	1.954	0.066	0.736	0.471	0.162	0.121	2.076	0.050
Largest basal diameter	0.380	0.045	0.737	<0.001	0.034	0.442	0.217	0.172
Thickness	0.190	0.204	0.654	<0.001	-0.153	0.254	0.385	0.042
Superior location	0.730	0.474	0.063	0.951	0.789	0.439	-0.296	0.771
Temporal location	0.809	0.428	-0.109	0.457	-0.729	0.475	-0.154	0.879
Nasal location	0.009	0.993	0.053	0.958	-0.169	0.867	0.621	0.541
Juxtapapillary location	0.936	0.361	1.320	0.202	0.059	0.953	1.617	0.122
Pigment	-0.972	0.343	-1.139	0.269	-0.896	0.381	-0.935	0.362

P = P value; r = Pearson correlation coefficient.

thickness ($P = 0.042$), advanced patient age ($P = 0.047$), and female gender ($P = 0.050$).

Adjuvant TTT

For the 9 patients treated with adjuvant TTT combined with plaque radiotherapy, the mean pretreatment visual acuity was 20/89 (range, 20/20 to 20/400), and the mean final visual acuity was 20/1045 (range, 20/25 to counting fingers). The mean change in lines of visual acuity was -1.9 lines (range, -9 to +5 lines). The tumor location was juxtapapillary (2 patients), juxtafoveal or subfoveal (4 patients), or juxtamacular (3 patients). The fundus quadrant involved was superior (4 patients), inferior (3 patients), nasal (one patient), or temporal (one patient). The mean largest basal tumor diameter was 10.4 mm (range, 5–16 mm), and the mean tumor thickness was 4.2 mm (range, 1.8–7.4 mm). Transpupillary thermotherapy was performed as a supplemental therapy in tumors that were unresponsive to plaque radiotherapy (6 patients) or as a prophylactic adjuvant treatment in tumors at high risk for local failure with plaque radiotherapy alone (3 patients). The mean number of TTT sessions was 1.1 (range, 1–2). The mean number of TTT spots per session was 13 (range, 6–20), and the mean total number of TTT spots was 29.6 (range, 6–26). The mean maximum treatment power for each patient was 605 mW (range, 350–1500 mW). Complications were limited to branch retinal vein or artery occlusion in 2 patients (22%); none of these complications affected visual outcome. The tumor flattened by ultrasonography in 3 patients, with a mean time to flattening of 13 months (range, 4–27

months; Fig 4). Transpupillary thermotherapy induced tumor shrinkage and resolution of exudative retinal detachment in all 6 tumors that did not respond to plaque radiotherapy. After a mean follow-up of 32 months (range, 10–52 months), all 9 tumors regressed satisfactorily, with no local failures or enucleations. No patients developed metastatic disease.

Case-Matched Comparison of Primary TTT versus Plaque Radiotherapy

Eighteen patients who were initially treated with TTT alone were matched case by case with 18 patients who were initially treated with plaque radiotherapy (Table 3). The TTT and plaque radiotherapy groups did not differ significantly in age (mean, 65.8 years in the TTT group vs. 61.5 years in the plaque radiotherapy group; $P = 0.332$), gender (11 males vs. 13 males; $P = 0.542$), initial visual acuity (mean, 20/100 vs. 20/97; $P = 0.810$), tumor location (juxtapapillary in 5 TTT patients vs. 2 plaque radiotherapy patients, juxtamacular or submacular in 9 TTT patients vs. 11 plaque radiotherapy patients, and juxtamacular in 4 TTT patients vs. 5 plaque radiotherapy patients; $P = 0.478$), or length of follow-up (mean, 33 vs. 40 months; $P = 0.253$). The mean tumor thickness was significantly greater in the plaque radiotherapy group (3.5 mm) than the TTT group (2.7 mm; $P = 0.001$). The mean largest basal tumor diameter was greater in the plaque radiotherapy group (9.5 mm) than the TTT group (7.4 mm), but the difference did not achieve statistical significance ($P = 0.060$).

Figure 2. Persistent tumor creep after primary transpupillary thermotherapy. **A**, Pretreatment appearance of juxtapapillary tumor. **B**, Tumor appearance 22 months later, with a flattening and atrophy of the tumor and patchy residual pigmentation. **C**, After 38 months and 4 treatment sessions, flat tumor tissue continues to expand at the temporal and nasal margins. **D–F**, Fluorescein angiograms corresponding to clinical photographs A–C, respectively. Note that the extent of tumor creep is more apparent by angiography than by clinical examination. The eye eventually required enucleation.

Figure 3. Local tumor recurrence after primary transpupillary thermotherapy. **A**, Pretreatment appearance of juxtapapillary tumor, nasal to disc. **B**, After complete tumor response with tumor flattening and replacement by an atrophic scar, a suspicious area of thin, pinkish-tan, choroidal tissue near the temporal edge of the treatment scar was identified 50 months after the initial treatment (arrow). **C–F**, Fluorescein angiogram frames at the time of clinical photograph (B). Note the abnormal, fine choroidal vessels near the nasal and temporal edges of the treatment scar. The abnormal tissue had a spongy appearance by stereoscopic viewing and leaked in the late frames, consistent with recurrent melanoma (arrows).

Figure 4. Successful adjuvant transpupillary thermotherapy after plaque radiotherapy. **A**, Pretreatment appearance of a juxtapapillary tumor that overhangs the disc. There was no optic nerve invasion by clinical examination or magnetic resonance imaging. The patient refused enucleation. **B**, Tumor appearance 48 months after plaque radiotherapy followed 3 months later by thermotherapy. Note the marked regression and color change within the tumor. **C**, Pretreatment fluorescein angiography, demonstrating a central collar-button break through Bruch's membrane. **D**, Fluorescein angiography 8 months after therapy, showing rapid regression of the collar button and involution of tumor blood supply. **E**, Pretreatment B-scan ultrasonography demonstrating a juxtapapillary, elevated tumor. **F**, B-scan ultrasonography 48 months after therapy, with flattening of the tumor and no extrascleral extension or optic nerve invasion.

Table 3. Case-Matched Comparison of Transpupillary Thermotherapy Versus Plaque Radiotherapy

Patient No.	Age (yrs)	Sex	LBD (mm)	Thickness (mm)	Location*	Initial Visual Acuity	Final Visual Acuity	Change in Acuity (Lines)	Local Failure	Metastasis	Follow-up (mos)
Transpupillary thermotherapy patients											
1	36	M	8	2.6	3	20/15	20/20	0	+	—	57
2	62	M	9	3.4	4	20/20	20/30	−2	—	—	40
3	82	F	6	2.8	2	20/25	20/50	−3	—	—	36
4	69	M	5	1.9	4	20/25	20/25	0	—	—	8
5	69	F	9	3.0	4	20/25	20/80	−6	—	—	32
6	63	F	8	3.1	3	20/25	20/25	0	—	—	58
7	63	F	6	3.0	2	20/40	20/400	−7	—	—	39
8	53	F	11	2.3	2	20/40	20/800	−8	—	—	47
9	54	M	6	1.8	1	20/50	20/25	−3	—	—	30
10	73	M	4	2.2	2	20/50	20/30	+2	—	—	4
11	73	F	4	2.1	3	20/50	Enucleated	−9	+	—	52
12	70	M	5	2.4	2	20/70	20/400	−4	—	—	12
13	79	M	5	1.9	1	20/80	20/4000	−4	—	—	28
14	78	M	10	4.2	2	20/100	Enucleated	−5	+	—	28
15	80	F	11	3.1	2	20/200	HM	−3	—	—	25
16	57	F	12	2.8	5	20/200	20/2000	−2	—	—	30
17	67	M	5	2.9	5	20/400	20/400	0	—	—	37
18	57	M	10	2.5	2	20/400	20/300	+1	—	—	34
Plaque radiotherapy patients											
1	63	F	10	3.9	1	20/20	20/50	−4	—	—	57
2	54	F	9.5	3.5	4	20/20	20/20	0	—	—	23
3	67	M	8.5	3.9	2	20/25	20/20	+1	—	—	26
4	38	F	12	4.0	3	20/25	CF	−10	—	—	62
5	65	M	11	3.1	3	20/30	20/600	−9	—	—	92
6	71	M	7	2.9	3	20/30	20/60	−3	—	—	26
7	78	M	12	4.4	2	20/40	20/25	+2	—	—	15
8	58	M	6	4.4	2	20/40	Enucleated	−10	+	—	4
9	69	M	8	2.5	1	20/50	20/1000	−7	—	—	27
10	38	F	11	3.4	5	20/50	20/30	+2	—	—	20
11	58	M	10	4.2	2	20/60	CF	−6	—	—	53
12	48	F	8	3.3	3	20/70	20/400	−4	—	—	82
13	64	M	6	2.9	2	20/80	HM	−5	—	—	58
14	77	F	12	2.6	2	20/80	20/2000	−4	—	—	38
15	59	M	7	3.2	3	20/100	20/400	−2	—	—	55
16	66	M	8	2.1	1	20/200	20/30	+7	—	—	21
17	60	M	12	4.3	2	20/400	HM	−2	—	—	39
18	77	M	9	4.3	5	20/400	20/2000	−1	—	—	38

CF = counting fingers; F = female; HM = hand movements; LBD = largest basal diameter; M = male.

*1 = juxtapapillary; 2 = macular (not subfoveal); 3 = juxtamacular; 4 = nasal; 5 = subfoveal.

The mean final visual acuity was 20/64 for the TTT group vs. 20/70 for the plaque radiotherapy group ($P = 0.810$). Approximately 2000 patients per group would be required to detect a statistically significant difference, assuming that all other variables remained unchanged. The mean posttreatment change in visual acuity lines was -2.9 lines for the TTT group vs. -3.1 lines for the plaque radiotherapy group ($P = 0.919$). The final visual acuity was 20/400 or worse in 9 (50%) TTT patients and 11 (61%) plaque radiotherapy patients (chi-square = 0.45; $P = 0.604$). Approximately 320 patients per group would be required to detect a statistically significant difference, assuming that all other variables remained unchanged. The final visual acuity was the same or better than the initial visual acuity in 6 (33%) TTT patients and 5 (28%) plaque radiotherapy patients (chi-square = 0.13; $P = 0.732$). There was no significant difference in the time to vision failure (visual acuity $\leq 20/400$) between the 2 groups (chi-square = 1.18; $P = 0.2769$). Local tumor recurrence was detected in 3 (17%) TTT patients and 1 (6%) plaque radiotherapy patient ($P = 0.331$).

Discussion

Initial visual outcomes and local tumor control rates with primary TTT for small choroidal melanomas were encouraging.^{8,10} However, these studies were published with limited follow-up, and they did not compare outcomes with those of accepted modalities such as plaque radiotherapy. Herein, we report the longest mean follow-up to date (32 months) for primary TTT. We observed retinal complications in 76% of patients and local treatment failure in 29% of patients. We also report the first case-matched comparison of primary TTT vs. plaque radiotherapy. Primary TTT did not result in better visual outcomes than plaque radiotherapy. We conclude that primary TTT for choroidal melanoma may not provide superior visual outcomes and has a higher risk of local failure than plaque radiotherapy. Therefore, the indications for TTT in the management of choroidal melanoma need to be re-examined.

Table 4. Summary of Published Reports of Transpupillary Thermotherapy Alone for Choroidal Melanoma

First Author	No. of Patients	Initial Mean Tumor Base (mm)	Initial Mean Tumor Thickness (mm)	Complications (%)	Local Failure (%)	Metastasis (%)	Follow-up, mos, Mean (Range)
Godfrey ⁸	14	6.8	1.8	NM	7	0	16 (7–28)
Robertson ⁹	20	5.5	1.8	75	1	0	18 (6–36)
Shields ¹⁰	17	6.6	3.0	NM	0	0	7 (6–10)
Shields ¹¹	256	7.1	2.7	64	9	1	19 (2–52)
Shields ¹²	100	7.1	2.8	39	6	0	14 (6–24)
This study	21	7.3	2.6	76	24	0	32 (4–58)

NM = not mentioned.

Oosterhuis et al² and Keunen et al⁴ originally developed TTT mainly as an adjuvant to plaque radiotherapy. However, in North America, TTT has been widely adopted as primary therapy for small choroidal melanomas.^{8,9,11} The indications for primary TTT in our study were similar to those in previous reports. Thin, posterior tumors near the macula or optic nerve were treated with the goal of avoiding radiation-related vision loss associated with plaque radiotherapy. We treated only lesions that were highly suggestive of choroidal melanoma on the basis of documented growth or the presence of at least 3 established risk factors for growth (subretinal fluid, orange pigmentation, thickness >2 mm, and visual symptoms).^{15,16} We performed TTT on some highly suspicious tumors near the fovea without waiting for growth, because extension into the fovea would preclude the possibility of retaining good vision. It is possible that some of these lesions were in fact suspicious nevi rather than small melanomas. Although a detailed discussion of treating suspicious nevi is beyond the scope of this article, the disappointing visual results with TTT suggest that the idea of treating suspicious nevi before growth may need to be re-evaluated until a better treatment option is available.

Retinal complications occurred in 76% of patients, and visual acuity declined by at least 2 lines in 57% of patients treated with primary TTT; these results are similar to those in previous studies.^{9,11} Visual acuity loss was associated with an increased total number of TTT spots ($P = 0.007$), increased spots per session ($P = 0.002$), increased number of TTT sessions ($P = 0.019$), and increased maximum TTT power ($P = 0.001$). Thus, heavier TTT treatments, which were often necessary to maintain local tumor control, were strongly associated with vision loss. Treatment of tumors located nasal to the optic disc may be less likely to result in vision loss, although several patients in our study with nasal tumors experienced significant vision loss as a result of massive epiretinal membrane formation that extended into the macula. Further studies are needed to determine whether some tumor locations may provide acceptable visual outcomes with primary TTT.

Two major goals of TTT are to flatten the tumor and to eliminate residual tumor pigmentation. However, tumor flattening (achieved in 81% of tumors) did not ensure local control, because 3 such tumors later recurred from areas of residual pigmentation. Elimination of residual pigmentation was difficult to achieve; persistent pigment within the treat-

ment scar was detected at the last office visit in 93% of patients. Further, it was not always possible to distinguish between inconsequential pigment clumping and residual tumor cells on clinical examination (Fig 2).

We were disappointed to observe a local failure rate of 29%. There are several potential explanations for this high failure rate compared with those in previous reports.^{9,11} First, the mean follow-up in this study was substantially longer than in previous reports, and there is a strong correlation between local failure and length of follow-up (Table 4). Second, our definition of local failure may have been more stringent than that in previous reports. Because of concern about potential metastatic risk, we had a low threshold for declaring local failure and proceeding with either plaque radiotherapy or enucleation. We recently reported a local failure rate of only 4% with plaque radiotherapy,¹⁷ and although those results cannot be compared directly with those from this study, the large discrepancy in failure rates between TTT and plaque radiotherapy is disconcerting. Although we did not observe any metastatic events in this study, we remain concerned that local failure after TTT may be a risk factor for metastatic disease, as it is after radiotherapy.^{18,19} Local failure was strongly associated with decreased tumor pigmentation ($P = 0.0001$), which has been recognized by other investigators and may be due to reduced conversion of infrared laser into heat within the tumor.³ Currently, we consider light tumor pigmentation as a relative contraindication to primary TTT.

The superior local control rates with plaque radiotherapy are likely due to several factors. First, radiotherapy induces widespread DNA damage, which kills many tumor cells and prevents surviving cells from proliferating.²⁰ Second, plaque radiotherapy is designed to deliver radiation beyond the detectable tumor margins, increasing the likelihood of eradicating occult tumor cells beyond the tumor margins. In contrast, TTT induces acute cytotoxic changes within tumor cells but is not known to have long-lasting effects on tumor cells that survive treatment.²¹ The heat from TTT dissipates within the choroid just beyond the tumor margins, where occult tumor cells may survive treatment. In addition, TTT is not effective in eradicating intrascleral tumor cells.⁴

Two forms of local treatment failure were noted after TTT. The first form was persistent tumor creep, characterized by persistence and expansion of tumor tissue (usually pigmented) at one or more tumor margins despite flattening of the tumor (Fig 2). Heavy retreatment of these tumors did

not prevent further tumor creep, suggesting intrinsic tumor resistance to thermotherapy. The second form of failure was local tumor recurrence, which was characterized by an initial complete response (tumor flattening and replacement of the tumor with an atrophic scar) followed by recurrence of vascularized tumor tissue, usually at an edge of the treatment scar (Fig 3). Fluorescein angiography was helpful in detecting local failures at an early stage (Figs 2, 3), and we perform angiography at regular follow-up intervals.

Because a major rationale for primary TTT is to avoid radiation-induced vision loss, we performed a case-matched comparative analysis of patients treated with primary TTT vs. similar patients treated with plaque radiotherapy. The only significant baseline difference between the 2 groups was a larger mean tumor thickness in the plaque radiotherapy group; this would tend to bias the study in favor of TTT. Nevertheless, there was no significant difference in visual outcomes between the 2 treatment groups with a mean follow-up of 40 months for the plaque radiotherapy group. There was no time-dependent difference in the vision loss between groups, although it is possible that a visual acuity difference could be detected with longer follow-up because of radiation retinopathy in the plaque radiotherapy patients. Limitations of this analysis include retrospective data collection, a small number of patients, and limited follow-up.

We previously showed that tumors that fail to respond to plaque radiotherapy with shrinkage and resolution of exudative detachment within 9 to 12 months after radiotherapy are likely to fail treatment,¹⁷ so we use adjuvant TTT in this setting. We also used adjuvant TTT after plaque radiotherapy in patients who were thought to be at high risk for local recurrence because of juxtapapillary tumor location or intraoperative ultrasonographic detection of plaque tilting away from the sclera.²² In contrast to primary TTT, adjuvant TTT after plaque radiotherapy rapidly induced resolution of exudative retinal detachment and regression of the tumor. Local control was achieved with only one TTT session in most patients. Similar encouraging results with adjuvant TTT have been reported by centers in Europe and North America.^{2,3,6,7,23}

In summary, this study suggests that primary TTT is unlikely to result in better visual outcomes and is more likely to result in local failure than plaque radiotherapy. Therefore, the indications for primary TTT may be more limited than initially anticipated. Presently, the most promising role for TTT is as an adjunct to plaque radiotherapy. Additional studies are needed to determine the indications and contraindications for primary and adjuvant TTT.

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